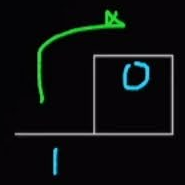
Staircase Problem

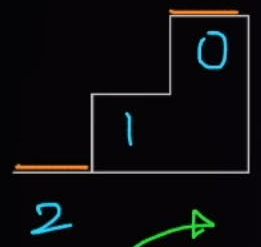
N = 0

1 <- { [0] }



N = 1

1 <- { [1, 0] }

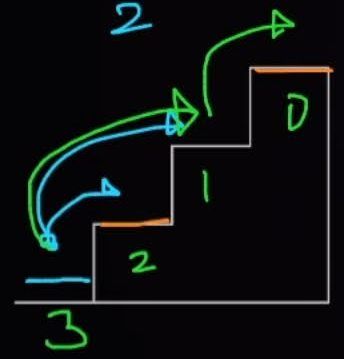


N = 2

2 <- { [2, 1, 0], [2, 0] }

n n-1 n-2

num_ways(3) = num_ways(2) + num_ways(1)



N = 3

3 <- { [3, 2, 1, 0], [3, 2, 0], [3, 1, 0] }

N = 4

5 <- { [4, 3, 2, 1, 0], [4, 3, 2, 0], [4, 3, 1, 0], [4, 2, 1, 0], [4, 2, 0] }

Ad in 2

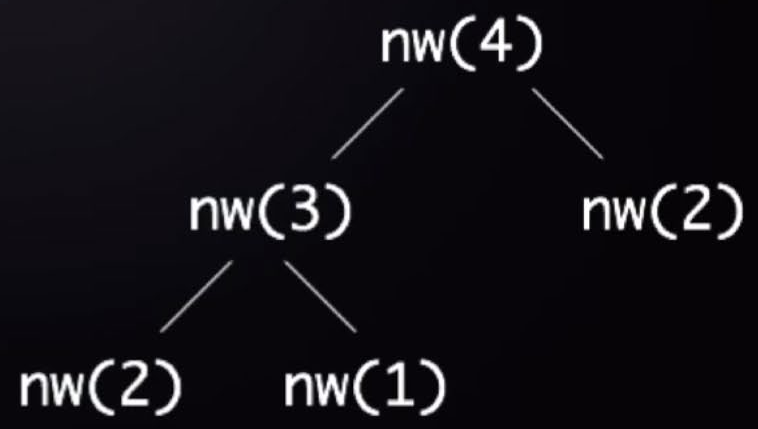
N	num_ways(N)
0	1
1	1
2	2
3	3
4	5

num_ways(N) = num_ways(N-1) + num_ways(N-2)

num_ways(0) = 1

num_ways(1) = 1

```
def num_ways(n):
    if n == 0 or n == 1:
        return 1
    else:
        return num_ways(n-1) + num_ways(n-2)
```



DP

N	num_ways(N)
0	1
1	1
2	2
3	3
4	5

num_ways(N) = num_ways(N-1) + num_ways(N-2)

num_ways(0) = 1

num_ways(1) = 1

```
def num_ways(n):
    if n == 0 or n == 1:
        return 1
    else:
        return num_ways(n-1) + num_ways(n-2)
```

```
def num_ways_bottom_up(n):
    if n == 0 or n == 1: return 1
    nums = new int[n+1]
    nums[0] = 1; nums[1] = 1
    for i from 2 upto n:
        nums[i] = nums[i-1] + nums[i-2]
    return nums[n]
```